

Random Subset Methods for Forecast Combination

Theory · Monte Carlo Simulations · Empirical Results — Alignment Report

4 June 2026

This document summarises the current state of the RSM project across three components: (i) theoretical results derived in the proposal and addendum, (ii) exogenous and endogenous Monte Carlo simulations, and (iii) the GDP empirical application using FRED-MD/FRED-QD real-time vintages. Each theoretical prediction is evaluated against its simulation and empirical counterpart.

Theory

Model

The common-loading model specifies the joint error covariance as $\Sigma_e = q\iota_m\iota_m' + D$, $D = \text{diag}(\sigma_1^2, \dots, \sigma_m^2)$, precision weights $\tau_i = \sigma_i^{-2}$. Two population risk objects are central:

Object	Formula	Interpretation
Q_k^{sub}	$q + \mathbb{E}_S[1/A_S]$	Average-subset risk (B subsets averaged)
$Q_k^{\text{bag}, \infty}$	$\bar{v}_k' \Sigma_e \bar{v}_k$	Infinite-bagging risk ($G \rightarrow \infty$)
$Q_k^{\text{bag}, G}$	$Q_k^{\text{bag}, \infty} + G^{-1}(Q_k^{\text{sub}} - Q_k^{\text{bag}, \infty})$	Finite-bagging bridge — exact identity

Key Theoretical Results

Result	Formula	Assumption
Small-het excess risk	$R_k^{\text{bag}, \infty} \approx C_\delta z_k^2$	$\tau_i \approx \bar{\tau}(1 + \delta_i)$, δ_i small
Cube-root optimal k^* (bagged)	$k^* \approx (2C_\delta T/A)^{1/3}$	Small-het + $T \gg k + z_k \approx 1/k$
Square-root optimal k^* (subset)	$k^* \approx (A_0 T/B)^{1/2}$	Leading-order average-subset loss
Finite-bagging bridge	$Q_k^{\text{bag}, G} = Q_k^{\text{bag}, \infty} + G^{-1}(Q_k^{\text{sub}} - Q_k^{\text{bag}, \infty})$	Exact algebraic identity
RSM dominance (κ)	$Q_k^{\text{bag}, \infty} < L^{\text{full OLS}}$ for $k \geq \kappa$	$\kappa = 2$ in all simulated DGPs

Power-Law Generalisation (Addendum, Section 13)

The small-het formula underestimates the true excess risk. The addendum proposes $R_k^{\text{bag}, \infty} \approx C \cdot z_k^\alpha$ and establishes two limit cases:

Regime	True form of R_k	Slope $\hat{\alpha}$	k^* scaling
Homoskedastic	0	—	any k optimal
Small heterogeneity	$C_\delta z_k^2$	2 (exact)	$T^{1/3}$
Intermediate	Mixture	$\alpha \in (1, 2)$	$T^{1/(\alpha+1)}$
Large het (dominant forecaster)	$C'(kz_k)^2$	≈ 1 (grid OLS)	$T^{1/2}$

Generalised optimal $k^* = (C\alpha T/A)^{1/(\alpha+1)}$. A general proof that $\hat{\alpha} \in [1, 2]$ for arbitrary $\{\tau_i\}$ **remains open** (noted explicitly in the addendum).

Exogenous Monte Carlo

Design: $m = 50$, $T = 240$, Σ_e **known** (exogenous weights). $N_{\text{rep}} = 500$, $G = 800$, $N_{\text{sub}} = 2500$. Four DGPs: $B \in \{3, 8\} \times H \in \{1.05, 6\}$ (H =heterogeneity ratio $\max/\min \sigma_i^2$). **Status:** **baseline complete; scaling_T_m complete; joint_B_H_grid running.**

Baseline Results ($m = 50, T = 240$)

DGP	k^* form.	k^* grid	k^* emp.	Gain vs eq.	Gap vs ora.	$\hat{\alpha}$	R^2
$B=3, H=1.05$	1	1	3	0.04%	0.19%	1.05	0.982
$B=8, H=1.05$	1	1	3	0.09%	0.45%	1.05	0.976
$B=3, H=6$	3	3	6	10.4%	0.92%	1.66	0.986
$B=8, H=6$	5	5	9	42.1%	3.23%	1.68	0.983

Cube-root formula matches population grid k^* exactly in all four DGPs. Empirical $k^* \approx 2 \times$ population k^* (Σ_e estimation noise penalty). Gains only meaningful when H is large.

Small-Het Approximation Quality

The ratio $R_k^{\text{mc}}/R_k^{\text{small-het}}$ grows with k and blows up near $k = m$. Counterintuitively, the approximation is **worst** for $H = 1.05$: the true curve follows $C''(kz_k)^2$ while the formula gives $C_\delta z_k^2$, diverging at rate k^2 .

DGP	Ratio at $k = 5$	Ratio at $k = 20$	Mean rel. error
$B=3, H=1.05$	3.9×	21×	91%
$B=8, H=1.05$	3.3×	16×	90%
$B=3, H=6$	2.6×	3.8×	72%
$B=8, H=6$	2.3×	9.3×	78%

Scaling Experiment ($T \in \{80, 240, 640, 1000\}, H = 1.05$)

The $H=1.05$ DGP has near-zero excess risk relative to $Q_{\text{opt}} \approx 1.06$, so population $k^*(\text{grid}) = 1$ for all T — the cube-root rule is not operational. The **H=6 scaling run** (500 reps, paper profile) is currently running.

T	k^* pop. grid	k^* cube-root cont.	k^* empirical
80	1	0.57	2
240	1	0.82	2
640	1	1.13	4
1000	1	1.31	3

Empirical log-log slope: 0.238 (p=0.21, not significant; k^* pinned near floor). $H=6$ run pending — expected to show clear cube-root exponent.

Endogenous Monte Carlo

Design: $m = 50, T = 200, \Sigma_e$ estimated from data (endogenous weights). Two error structures (Toeplitz $\rho = 0.8$, Factor $r = 2$) $\times$ two β designs (B2 dense, B4 sparse). $N_{\text{rep}} = 200, K_{\text{RS}} = 200$, OOS=30. **Status:** complete.

Four-Scenario Summary

Scenario	k^* RMSFE	Rel. RMSFE	Gain	$k^*(L_{\text{emp}})$	κ
Toeplitz $\rho = 0.8$, B2 (dense)	11	0.965	−3.5%	32	2
Toeplitz $\rho = 0.8$, B4 (sparse)	11	0.902	−9.8%	26	2
Factor $r = 2$, B2 (dense)	14	0.950	−5.0%	29	2
Factor $r = 2$, B4 (sparse)	14	0.939	−6.1%	29	2

$\kappa = 2$ universally: bagged RSM beats full estimated OLS for any $k \geq 2$. Sparsity (B4) amplifies gains. Factor structure shifts k^* : $11 \rightarrow 14$.

T-Scaling of Empirical k^* (B4, 120 reps, $T \in \{100, \dots, 800\}$)

Σ_e type	100	200	300	500	800	Log-log slope	Slope = $\frac{1}{3}$?	Slope = $\frac{1}{2}$?
Toeplitz	10	14	14	18	22	0.394**	Cannot reject	Rejected ($t = -2.1$)
Factor	14	14	18	26	26	0.367**	Cannot reject	Rejected ($t = -1.9$)

The log-log slope (0.37–0.39) is statistically indistinguishable from $\frac{1}{3}$ but significantly below $\frac{1}{2}$. The appearance that $k^* \approx \sqrt{T}$ at $T = 100$ –200 is a *coincidence*: $c \cdot T^{1/3} = \sqrt{T}$ exactly when $T = c^6 \approx 113$ (with $c \approx 2.2$). At $T = 800$: cube-root gives ≈ 20 (matches data=22); $\sqrt{T} = 28.3$ (does not).

Population vs Empirical k^* Gap

Criterion	Loss surface	k^* (Toeplitz B2)
L_{pop} (known Σ_e)	Range 0.009 — nearly flat	2
RMSFE (OOS, empirical)	—	11
L_{emp} (estimated $\hat{\Sigma}_e$)	Range 0.39 — steep	32

The inflation formula $(T-1)/(T-k)$ accounts for estimating k combination weights given fixed Σ_e . It does **not** account for estimating Σ_e from $m = 50$ series — this additional noise pushes empirical k^* from 2 to 11.

GDP Empirical Application (FRED-MD/FRED-QD)

102 real-time vintages (1999Q3–2025Q1, vintage 18 excluded). Predictor pool: $m \approx 126$ monthly FRED-MD series, each transformed to a predicted quarterly growth rate by a rolling univariate AR ($n_{\text{start}} = 40$ burn-in quarters). k -grid: 2–55. RSM draws: $B_{\text{main}} = 1000$, $B_{\text{allk}} = 200$. **Status: complete** (minor legend bug in combined figure; all data saved).

Sample Design: Why $T \approx 81$ and $m \leq 55$

Parameter	Value	Source / explanation
Training obs T	79–180 (median 81)	FRED-QD starts $\approx 1970Q1$; after $n_{\text{start}} = 40$ burn-in: $T \approx T_{\text{QD}} - 40$
Rule for k_{main}	$\text{round}(\sqrt{T}) = 9$ (median)	Hard-coded heuristic; cube-root formula not used
Predictor pool m	≈ 126 FRED-MD series	Monthly-to-quarterly aggregation; each column is a univariate AR forecast of that series
k search range	2 to 55	Hard-coded cap; at $k = 55$, $T = 81$: $k/T \approx 0.68$, inflation factor $(T - 1)/(T - k) \approx 3.1$
C_δ estimated?	No	$m > T$ makes Σ_e estimation infeasible; C_δ , α , A all unobservable

Forecast Accuracy Tables**Table 1 — Full Sample**

Model	MAFE	Sig.	RMSF	Sig.	Target MAFE	Target RMSFE
Mean	0.54		1.10		0.54	1.10
RSM	0.45	**	0.76	*	0.45	0.74
Lasso	0.51		0.73	*	0.51	0.76
Ridge	0.49	*	0.98		0.50	1.00
Random Forest	0.55		1.24		0.55	1.24

Table 2 — COVID-19 Excluded

Model	MAFE	Sig.	RMSF	Sig.	Target MAFE	Target RMSFE
Mean	0.42		0.60		0.42	0.60
RSM	0.38	**	0.53	**	0.39	0.53
Lasso	0.45		0.58		0.45	0.60
Ridge	0.39	**	0.54	**	0.39	0.55
Random Forest	0.41		0.57	*	0.41	0.58

RSM matches paper targets within ± 0.02 RMSFE. Minor discrepancies reflect vintage coverage differences and random seeds.

Optimal k vs Theory Benchmarks ($T \approx 81$)

Benchmark	Full sample	No-COVID
k^* (RMSFE, actual)	19	21
$\sqrt{T}$ with $T = 81$	9.0	9.0
$T^{1/3}$ with $T = 81$	4.3	4.3
$2.2 \times T^{1/3}$ (endo. Toeplitz constant)	9.5	9.5
$2.8 \times T^{1/3}$ (endo. Factor constant)	12.1	12.1
Power-law slope $\hat{\alpha}$ ($k = 2 \dots 30$)	0.10	0.04

Empirical $k^* = 19\text{--}21$ is $\approx 2\times$ larger than the endogenous simulation predicts at the same T . The RMSFE curve is nearly flat ($\hat{\alpha} \approx 0.04\text{--}0.10$, far below the theoretical range $[1, 2]$); with only 102 OOS predictions the curve is estimated with high sampling variance. The $\sqrt{T}$ heuristic ($k = 9$) costs only 0.09 RMSFE relative to the empirical optimum in the full sample.

Theory–Simulation–Empirical Alignment

Each row is a theoretical prediction; columns show whether it is **confirmed** (green), **partial** (amber), **not confirmed** (red), or — N/A in each component.

Theoretical Prediction	Exogenous MC	Endogenous MC	GDP Empirical
Finite-bagging bridge identity	✓ Confirmed $\max \text{err} < 9 \times 10^{-15}$	— N/A	— N/A
Cube-root formula = population k^*	✓ Exact (all DGPs)	~ Partial Pop. $k^* = 2$; emp. $k^* = 11$	✗ Not applicable C not estimable ($m > T$)
Square-root subset formula = grid k^*	✓ Exact (all DGPs)	— N/A	— N/A
Empirical k^* scales as $T^{1/3}$	~ Partial $H = 1.05$: k^* at floor; $H=6$ run pending	✓ Confirmed Slope 0.37–0.39 ($p < 0.01$)	~ Partial k^* grows; curve too flat
Power-law $\hat{\alpha} \in [1, 2]$, $R^2 > 0.97$	✓ Confirmed $\hat{\alpha} \in \{1.05, 1.68\}$	— N/A	✗ $\hat{\alpha} \approx 0.04\text{--}0.10$ Sampling noise dominates
$\hat{\alpha}$ depends on H , not B	✓ Confirmed $B=3$ and $B=8$ identical	— N/A	— N/A
RSM dominates full estimated OLS	✓ $\approx 25\%$ lower loss	✓ $\kappa = 2$ universally	✓ OLS infeasible; RSM works
RSM beats equal weights (large H)	✓ 10–42% gains	✓ 3.5–9.8% RMSFE	✓ 5–12% RMSFE gain
Small-het approximation accurate	✗ Off by 2–21$\times$ 72–91% rel. error	— N/A	— N/A
$k^* \approx \sqrt{T}$ as rule of thumb	~ Coincidence at $T \approx 113$	~ Valid at $T = 100\text{--}200$ only; true scaling is $T^{1/3}$	~ Used in script; grid k^* is $\approx 2\times$ larger

Open Items and Next Steps

Priority	Item	Status
High	H=6 scaling run (500 reps, paper profile): verify cube-root exponent $\approx 1/3$ for exogenous case with meaningful heterogeneity	Running — PID active, $\approx 4\text{--}5$ h remaining
High	joint_B_H_grid: full $\hat{\alpha}(H)$ surface, $B \times H$ interaction, cube-root scaling at $H = 6$	Running — begins after <code>scaling_T_m</code>
High	Prove $\hat{\alpha} \in [1, 2]$ for arbitrary $\{\tau_i\}$: upper bound (Jensen) plausible; lower bound needs monotonicity in H or per-forecaster decomposition	Open theoretical problem
High	Explain robustness of cube-root formula: why $k^*(\text{formula}) = k^*(\text{grid})$ when small-het approx fails by $3\text{--}21\times$?	Empirical finding — no proof yet
Medium	Endogenous G_bag fix: re-run with <code>G_bag=K_RS</code> to populate <code>Q_bag_G_mc</code> column (currently all NA)	Bug identified, not yet re-run
Medium	GDP legend fix: one-line <code>pt.pch</code> removal to regenerate combined figure	Trivial — pending
Medium	Integrate addendum (Section 13) into main proposal: decide placement (Sec. 7 extension vs standalone appendix)	Decision pending
Medium	Formalise practitioner rule of thumb: $k^* \in [T^{1/3}, T^{1/2}]$ as a documented recommendation	Needs paper section
Low	Fix GDP script bugs for final results: <code>exclude=1845, ak[-c(1:2),], dm.test</code> NAs	Identified, deferred

Generated 4 June 2026. Exogenous PID 71717 (joint_B_H_grid running). H=6 scaling PID active (500 reps, paper profile). GDP outputs: GDP_constrained/all_k_results/. LaTeX addendum: rsm_powerlaw_section.tex.